- What speed has to be applied to fire a satellite upward so that it never returns to the earth?
- What is the radius of the smallest circular disk that can cover every isosceles triangle of given perimeter P?
- What volume of material is removed from a solid sphere of radius $2r$ if a hole of radius r is drilled through the centre?
- If a strain of bacteria grows at a rate proportional to the amount present and if the population doubles in one hour, how much will it increase at the end of two hours?

Integration will answer for all the above problems.

Learning Objectives

On completion of this chapter, the students are expected to

- understand the definition of an indefinite integral as a result of reversing the process of differentiation
- find the indefinite integrals of sums, differences and constant multiples of certain elementary functions.
- use the appropriate techniques to find the indefinite integrals of composite functions.
- apply integration to find the function, when the rate of change of function is given.

11.2 Newton-Leibnitz Integral

Integral calculus is mainly divided into indefinite integrals and definite integrals. In this chapter, we study indefinite integration, the process of obtaining a function from its derivative.

We are already familiar with inverse operations. $(+,-)$, $(\times,\div)$, $\left((\,)^n, \sqrt[n]{\ }\right)$ are some pairs of inverse operations. Similarly differentiation and integrations $(d, \int)$ are also inverse operations. In this section we develop the inverse operation of differentiation called 'antidifferentiation'.

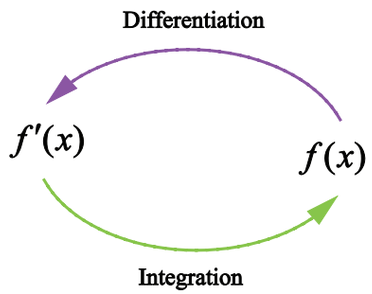


Fig. 11.4

Definition 11.1

A function $F(x)$ is called an **antiderivative** (Newton-Leibnitz integral or primitive) of a function $f(x)$ on an interval I if

$$F'(x) = f(x), \text{ for every value of } x \text{ in } I$$

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